

THE EFFECT OF BIAS CONDITIONS ON ALGaN/GaN 2DEG HALL PLATES

Karen M. Dowling^{*1}, Hannah S. Alpert¹, Pengxiang Zhang², Andrea N. Ramirez¹, Ananth Saran Yalamathy¹, Helmut Köck³, Udo Ausserlechner³, and Debbie G. Senesky¹
¹Stanford University, Stanford, California, USA, ²Tsinghua University, China, and ³Infineon Technologies AG, Villach, Austria

ABSTRACT

This paper describes the operation of AlGaIn/GaN two-dimensional electron gas (2DEG) Hall plates under various supply conditions (0.026 V to 1.27 V). The 100- μm -diameter octagon-shaped devices were microfabricated using metal-organic chemical vapor deposition of AlGaIn/GaN on <111> silicon wafers and traditional photolithography techniques. Upon device characterization at various Hall supply voltages, we observed an increase in the residual offset from 0.1 mT to 1.4 mT (from 9% of measured signal to over 60% in a 1 mT magnetic field). In addition, the sensitivity (scaled with bias voltage) was constant at 76 ± 2.5 mV/V/T (stable within 3%) with high linearity ($R^2 > 0.99$) across the tested operating conditions. This work demonstrates improved understanding of AlGaIn/GaN sensor elements that may be monolithically integrated with power electronics, as well function within extreme environments.

INTRODUCTION

Hall-effect devices are used in diverse sensing applications such as position and velocity sensors in automobiles and current sensing in power electronics [1]. However, silicon-based Hall-effect sensors have limitations in extreme environments because of the influence of temperature on intrinsic carrier concentration – low-doped materials ($< 10^{16} \text{ cm}^{-3}$) become saturated with carriers around 300°C. Recently, two-dimensional electron gas (2DEG) material systems, including AlGaIn/GaN, have gained high interest for power electronics monitoring and extreme environment sensing due to their durable nature, wide bandgap, and potential for monolithic integration with electronics [2]. Additionally, piezoelectric and spontaneous polarization create a stable 2DEG carrier concentration across a wide temperature range [3], which enables robust Hall-effect sensing [4], [5]. However, sensing low magnetic field signatures (< 10 mT) under high bias conditions has not been investigated with AlGaIn/GaN 2DEG Hall plates.

The Hall-effect can be leveraged for sensing magnetic field signatures through a 4-probe scheme. Constant current is applied across two contacts, and the induced electric potential from the external magnetic field is measured by a Hall voltage reading on the other two contacts. AlGaIn/GaN 2DEG plates are promising candidates for Hall-effect sensing because of their high mobility ($\sim 2200 \text{ cm}^2/\text{V}\cdot\text{s}$) [4] and high temperature stability.

In applications with low magnetic field levels, sensors are required to have high signal accuracy to overcome issues with background fields such as Earth's field or from electromagnetic interference. Hall devices generally have high raw offset values (larger than Earth's magnetic field of 50 μT) when no external magnetic field is applied due to inherent material defects or from various steps in microfabrication [6], [7]. This limits the minimum detectable signal to inconvenient ranges and reduces sensor accuracy. To overcome this issue, current spinning [7] and orthogonal layouts [8] have been adopted in practice to remove these “raw” offsets for silicon devices. These techniques can reduce the raw offset to a “residual” offset below 10 μT under various conditions [6], [7], [9], [10], corresponding to an improvement by a factor of up to 4,000.

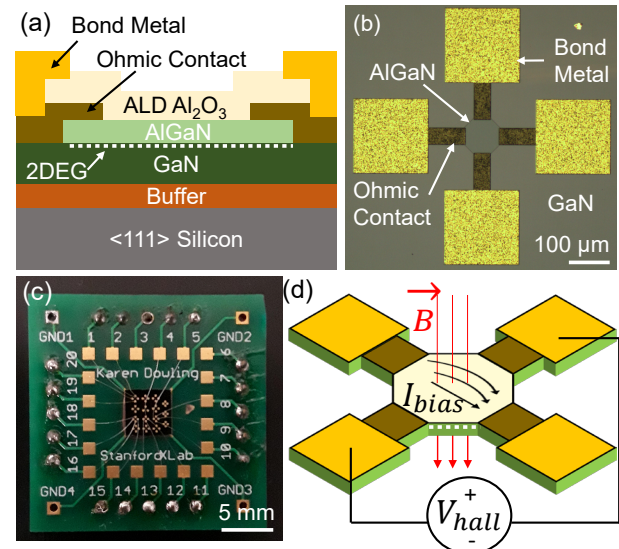


Figure 1: (a) Cross sectional schematic of the AlGaIn/GaN 2DEG Hall plate. (b) Optical image of the 2DEG Hall plate with 4 contacts (no top plate). (c) Packaged Sensor #1 for testing with floating substrate. (d) Device operation. Constant current is applied across a 4-contact van der Pauw structure and the Hall voltage is measured across the other two contacts.

In addition to low residual offset values, sensors are also required to have high sensitivity in many applications. Sensitivity is defined as a ratio of the change in Hall voltage (V_{Hall}) due to an external magnetic field (B). It is well known that the Hall voltage scales with supply voltage (or current), so most reports show either sensitivity scaled with supply current (S_I) or sensitivity scaled with supply voltage (S_V). However, in most applications, Hall plates are operated with a constant supply voltage to enable ease of integration with interface circuitry.

In this paper, we present the methodology for Hall-effect sensor microfabrication and characterization with a current spinning technique to study offset values in low magnetic fields (< 5 mT). We also examine the sensitivity and residual offset in AlGaIn/GaN 2DEG Hall plates as influenced by supply voltage. We observed that the sensor exhibited large residual offset values (60% of the signal in a 1 mT field) under high supply voltage conditions, as well as a constant sensitivity of 76 ± 2.5 mV/V/T. A discussion on the leading contributions to residual offset in these high bias schemes is provided. It should be noted that the AlGaIn/GaN Hall-effect sensors have higher sensitivity values compared to silicon-based Hall plates and with proper offset calibration can be used over a large temperature range for extreme environments.

METHODOLOGY

Fabrication

To study the effect of bias conditions on AlGaIn/GaN 2DEG Hall plate performance, we examined two devices (4-contact van der